

Lemma. Let $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ be a polynomial of degree n . Then, for each e , $0 < e < 1$, there exists an $R > 0$ such that for $|z| > R$

$$(1-e) |a_n| |z|^n < |p(z)| < (1+e) |a_n| |z|^n$$

Let r be rational function, $r: \mathbb{R}_\infty \rightarrow \mathbb{R}_\infty$. For $r = \frac{p}{q}$, where p and q have no common zeros, define

$\deg(r) = \deg(p) - \deg(q)$ and let pole set of r , P_r , be given as $P_r = Z_q$.

Theorem. Let r be rational function, $r: \mathbb{R}_\infty \rightarrow \mathbb{R}_\infty$. Suppose

(1) $\deg(r) \leq -2$ and

(2) $P_r \cap \mathbb{R} = \emptyset$.

Then,

$$\int_{-\infty}^{\infty} r(x) dx = 2\pi i \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \text{Res}(r(z); a_k)$$

Corollary. Suppose additionally, r is even. Then,

$$\int_0^{\infty} r(x) dx = \pi i \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \text{Res}(r(z); a_k)$$

Examples. $\int_{-\infty}^{\infty} \frac{x+1}{x^4+1} dx, \int_0^{\infty} \frac{x^2+1}{x^4+1} dx,$

Exceptions. $\int_0^{\infty} \frac{x}{x^4+1} dx, \int_0^{\infty} \frac{1}{x^3+1} dx$

Theorem. Let r be rational function, $r : \mathbb{R}_\infty \rightarrow \mathbb{R}_\infty$. Suppose

- (1) $\deg(r) \leq -1$ and
- (2) $P_r \cap \mathbb{R} = \emptyset$.

Then,

$$\int_{-\infty}^{\infty} \cos(x) r(x) dx = \operatorname{Re} \left(2\pi i \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \operatorname{Res}(e^{iz} r(z); a_k) \right) \text{ and}$$

$$\int_{-\infty}^{\infty} \sin(x) r(x) dx = \operatorname{Im} \left(2\pi i \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \operatorname{Res}(e^{iz} r(z); a_k) \right)$$

Corollary. Suppose additionally, r is even. Then,

$$\int_0^{\infty} \cos(x) r(x) dx = \operatorname{Re} \left(\pi i \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \operatorname{Res}(e^{iz} r(z); a_k) \right)$$

Alternately, suppose additionally, r is odd. Then,

$$\int_0^{\infty} \sin(x) r(x) dx = \operatorname{Im} \left(\pi i \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \operatorname{Res}(e^{iz} r(z); a_k) \right)$$

Examples. $\int_0^{\infty} \frac{x \sin(x)}{x^2 + a^2} dx, a \in \mathbb{R} \setminus \{0\}$

Theorem. Let r be rational function, $r : \mathbb{R}_\infty \rightarrow \mathbb{R}_\infty$. Suppose

- (1) $\deg(r) \leq -1$ and
- (2) $P_r \cap \mathbb{R} = \{0\}$
- (3) r has simple pole at 0

Then,

$$P.V. \int_{-\infty}^{\infty} \sin(x) r(x) dx = \operatorname{Im} \left(2pi \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \operatorname{Res}(e^{iz} r(z); a_k) + pi \operatorname{Res}(e^{iz} r(z); 0) \right)$$

Corollary. Suppose additionally r is odd. Then,

$$\int_0^{\infty} \sin(x) r(x) dx = \operatorname{Im} \left(pi \sum_{\substack{a_k \in UHP \\ a_k \in P_r}} \operatorname{Res}(e^{iz} r(z); a_k) + \frac{p}{2} i \operatorname{Res}(e^{iz} r(z); 0) \right)$$

Example. $\int_0^{\infty} \frac{\sin(x)}{x} dx$

Extended Example. $\int_0^{\infty} \frac{\sin^2(x)}{x^2} dx$

Theorem. Let r be rational function, $r : \mathbb{R}_{\infty} \rightarrow \mathbb{R}_{\infty}$. Suppose

- (1) $\deg(r) \leq -2$ and
- (2) $P_r \cap \{\mathbb{R}^+ \cup \{0\}\} = \emptyset$.

Suppose $0 < a < 1$. Then,

$$(1 - e^{2pia}) \int_0^{\infty} x^a r(x) dx = 2pi \sum_{a_k \in P_r} \operatorname{Res}(z^a r(z), a_k)$$

where $z^a = \exp(a \log_0 z)$ and $\log_0 z = \log |z| + i \arg_0 z$, $0 < \arg_0 z < 2p$.

Example. $\int_0^{\infty} \frac{x^a}{x^2 + 3x + 2} dx$, $0 < a < 1$

Extended Example. $\int_0^{\infty} \frac{\log x}{(1 + x^2)^2} dx$

Theorem. Let $R(x, y)$ be rational in x, y . Suppose $R(\cos(q), \sin(q))$ is continuous on $[0, 2p]$. Then,

$$\int_0^{2p} R(\cos(q), \sin(q)) dq = \int_{|z|=1} R\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right) \frac{dz}{iz}$$

Example. $\int_0^{2p} \frac{dq}{2 + \cos q}$